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Review

Mechanistic insights and basis for real-time monitoring and closed-loop feedback control in sonodynamic therapy for glioblastoma

Harshal A. Shah [a,b], Hasan Slika [a], Fnu Ruchika [a], Danielle Golub [b], Michael Schulder [b], Henry Brem [a,c,d,e], Amir Manbachi [a,e,f], Jordina Rincon-Torroella [a], Chetan Bettegowda [a], Pavlos Anastasiadis [g,h,i,j], Francesco Prada [k,l,m], Graeme F. Woodworth [g,h,i,j], Betty M. Tyler [a,*]

[a] *Department of Neurosurgery, The Johns Hopkins University School of Medicine, Baltimore, MD, USA*
[b] *Department of Neurosurgery, North Shore University Hospital/Donald and Barbara Zucker School of Medicine at Hofstra/Northwell, Manhasset, NY, USA*
[c] *Department of Ophthalmology, The Johns Hopkins University School of Medicine, Baltimore, MD, USA*
[d] *Department of Oncology, The Johns Hopkins University School of Medicine, Baltimore, MD, USA*
[e] *Department of Biomedical Engineering, The Johns Hopkins University School of Medicine, Baltimore, MD, USA*
[f] *HEPIUS Innovation Laboratory, The Johns Hopkins University School of Medicine, Baltimore, MD, USA*
[g] *Department of Neurosurgery, University of Maryland School of Medicine, Baltimore, MD, USA*
[h] *Marlene and Stewart Greenebaum Comprehensive Cancer Center, University of Maryland School of Medicine, Baltimore, MD, USA*
[i] *Department of Diagnostic Radiology and Nuclear Medicine, University of Maryland School of Medicine, Baltimore, MD, USA*
[j] *Fishell Department of Bioengineering, University of Maryland, College Park, MD, USA*
[k] *Acoustic Neuroimaging and Therapy Lab, Istituto Neurologico C. Besta, Milan, Italy*
[l] *Department of Neurological Surgery, University of Virginia, Charlottesville, VA, USA*
[m] *Focused Ultrasound Foundation, Charlottesville, VA, USA*

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ABSTRACT

Sonodynamic therapy (SDT) involves the administration of otherwise inactive agents that can be activated by acoustic energy ('sonosensitizers') to impart therapeutic effects. SDT is a treatment of significant clinical interest in glioblastoma, a highly aggressive brain tumor, due to the known uptake and conversion of the clinically approved fluorescence guided surgery agent, 5-aminolevulinic acid (5-ALA). Building evidence suggests acoustic energy may activate the converted product of 5-ALA, protoporphyrin IX. Despite ongoing clinical trials using 5-ALA-based SDT demonstrating treatment safety, feasibility, and potential efficacy, the precise underlying tumoricidal mechanisms of SDT remain unknown. Additionally, the ability to monitor SDT effects during treatments remains underexplored. Here we synthesize existing evidence regarding mechanisms behind the antitumoral effects of SDT, including various SDT agents studied in their capacity to generate reactive oxygen species that result in intrinsic apoptotic pathway activation, sonomechanical effects that result in cellular damage, pyrolytic and sonoluminescent reactions, and immunological activation. Additionally, we discuss the opportunities for *in situ, real-time* monitoring of SDT and related effects to enable safe, reproducible, and prescriptive treatments. Specifically, we explore the potential utility of magnetic resonance (MR) based monitoring tools including MR thermometry and MR acoustic radiation force imaging, and acoustic emissions feedback monitoring.

1. Introduction

Glioblastoma (GBM) is the most common and aggressive primary central nervous system (CNS) malignancy, with a median overall survival of approximately 15 months [1,2]. Current therapy consists of maximal, safe cytoreductive surgery and adjuvant chemoradiation [3–5]. Improvements in therapeutic efficacy are challenged by difficulties in delivering active agents across the blood-brain barrier (BBB) while minimizing off-target toxicity [6]. Focused ultrasound (FUS) has shown promise in its image-guided, spatially precise capabilities in providing safe, repeatable, and well-tolerated antitumoral effects [7–11].

* Correspondence to: Department of Neurosurgery, The Johns Hopkins University School of Medicine, 1800 Orleans St, Baltimore, MD 21287, USA.
E-mail address: btyler@jhmi.edu (B.M. Tyler).

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A variety of FUS applications are being developed for GBM treatment, including FUS-mediated BBB-opening [10]. Recently, a new, less understood approach using FUS-mediated activation of otherwise biologically inert and acoustically sensitive agents is being explored. This technique is termed sonodynamic therapy (SDT) [8–10,12,13]. SDT has evolved from the relatively more established and mechanistically understood photodynamic therapy (PDT), which involves the light-based excitation of light-sensitive agents via photostimulation to impart tumoricidal effects [14–16]. The use of SDT has not matured enough to yield sensitizer-device combinations that are approved for clinical use. However, promising preclinical investigations on the efficacy of SDT have prompted ongoing clinical investigations, which have yielded encouraging results regarding safety and tolerability [8,17–23]. As such, SDT has evolved as a promising adjunctive treatment, offering a targeted and non-invasive means of inducing tumoral cell death, and therefore is especially promising in cases where gross total tumor resection is infeasible [24]. Given the current experimental nature of SDT, several outstanding questions remain unanswered regarding the effects of SDT at the infiltrating margins of the tumor and its effects in conjunction with other established therapeutic modalities, such as radiotherapy and chemotherapy [24]. Additionally, the mechanisms of action of SDT are not completely understood, limiting treatment monitoring, prescriptive dosing, and clinical translation. A thorough understanding of the different mechanisms implicated in the anti-cancer effects of SDT in the context of GBM is required to design the appropriate monitoring strategies and create a holistic system that can automatically adjust FUS parameters based on real-time in-situ feedback in a closed-loop manner to maximize safety and efficacy (Fig. 1).

Although the predominating theories behind SDT's mechanisms of cytotoxic activity involve the generation of reactive oxygen species (ROS), several other molecular and cellular-scale mechanisms have been shown to contribute to the tumoricidal effects of SDT. Among these proposed mechanisms are pyrolytic reactions and mechanical, antiangiogenic, and immunomodulatory effects. In this review, we summarize the existing evidence supporting various mechanisms behind the therapeutic effects of SDT and describe common monitoring techniques used to assess *in-situ* ultrasound effects.

2. Overview of SDT mechanisms of action

2.1. Acoustic cavitation and sonomechanical effects

When exposed to acoustic pressure, fluid-filled structures undergo mechanical changes, resulting in the formation of gas pockets. These pockets may dynamically alternate between states of compression and rarefaction in response to sinusoidal ultrasound waves—a phenomenon termed acoustic cavitation (e.g., acoustically driven bubble activity) [25]. At low acoustic pressures, acoustically driven bubbles oscillate in a controlled manner (stable cavitation) and were shown to increase the membrane permeability of surrounding cells (Fig. 2) [26]. Conversely, at high acoustic pressures, inertial cavitation (e.g., the unstable oscillation and collapse of bubbles) releases energy that may lead to the creation of extreme temperatures, rapid pressure changes, and microstreaming (Fig. 2). In turn, microstreaming may physically rupture surrounding biological entities [27,28]. Consequently, knowledge of the pressure threshold that could induce inertial cavitation is critical for predicting the effects induced by acoustically driven bubbles. Apfel and Holland estimated the pressure threshold of inertial cavitation in plain water and demonstrated an increase with frequency [29]. Most studies on estimating the cavitation threshold in specific environments have assumed freely moving bubbles [30–32]. In biological systems, however, bubbles circulate within vessels or spaces, altering their physics [33]. Using theoretical and empirical approaches in gel phantoms, Sassaroli and Hynynen reported that the threshold of inertial cavitation is inversely proportional to vessel diameter so long as the diameter remains under 300 microns [34].

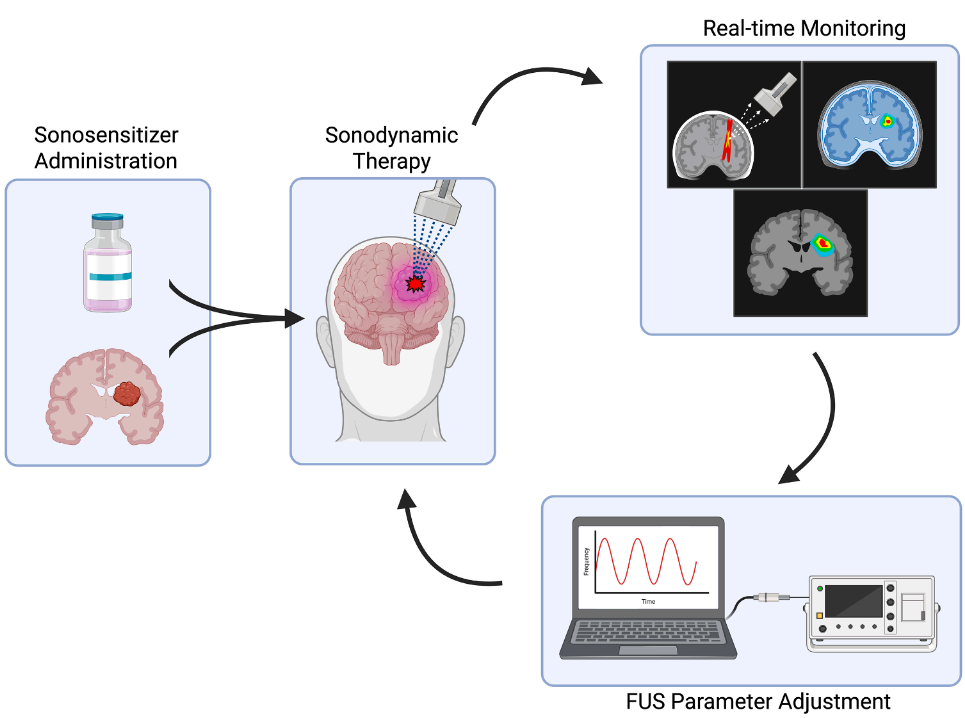

Fig. 1. Schematic overview of sonodynamic therapy for glioblastoma with a closed-loop feedback system. The procedure starts with the administration of a sonosensitizer that achieves good blood-brain barrier penetration and tumor accumulation, followed by the targeting of a focused ultrasound beam to the tumor area. Subsequently, in-situ real-time monitoring techniques (MR thermometry to monitor temperature changes and ensure no overheating, MR-ARFI to monitor acoustic intensity and guide accurate localization, and/or acoustic emission monitoring) are utilized. The monitoring of these parameters guides an automatic regulation of the focused ultrasound parameters whenever necessary, thus creating closed-loop feedback to enable a safe, precise, and effective procedure. ARFI, acoustic radiation force imaging; MR, magnetic resonance.

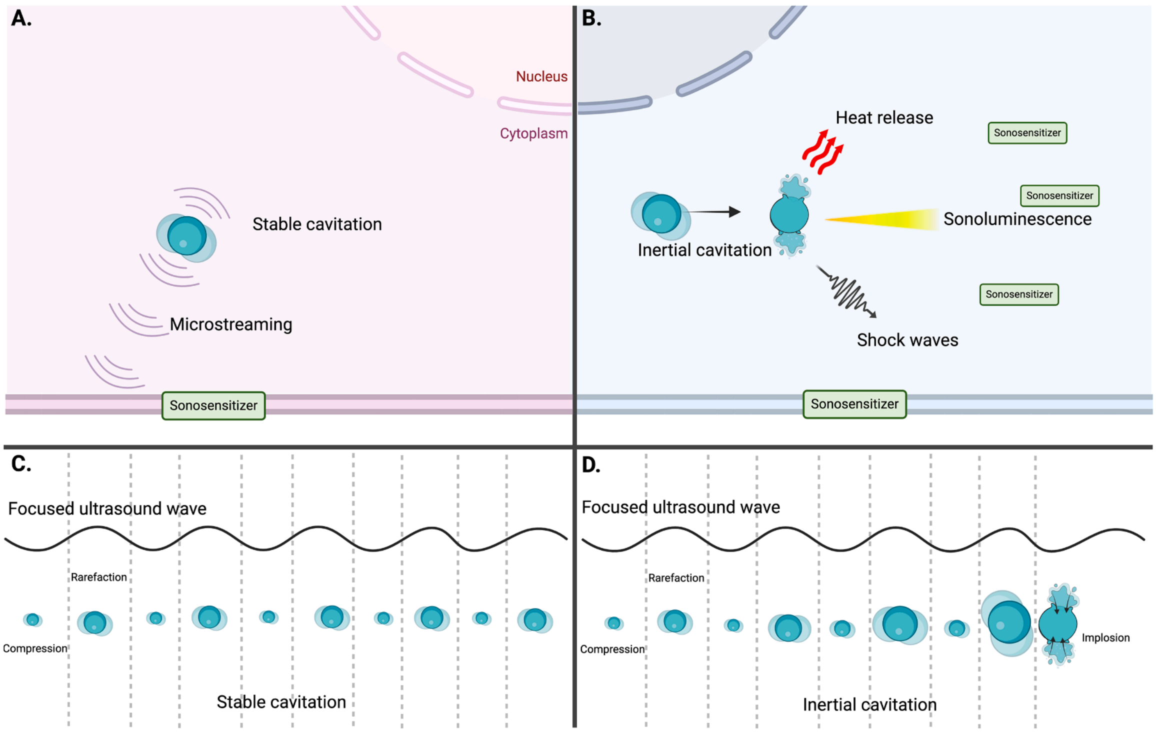

Fig. 2. The two types of acoustic cavitation are induced by focused ultrasound waves within cells. A. Stable cavitation of "cavitation nuclei" results in microstreaming which is transmitted to nearby structures and results in disruption of cell membranes, which may be predisposed to destabilization by a sonosensitizer agent. B. Inertial cavitation of the gas bubbles ultimately leads to their implosion, resulting in the release of substantial amounts of energy in the form of heat, sonoluminescence, and pressure waves. These lead to the transmission of energy to sonosensitizing agents and physiologic structures, mediating a myriad of mechanical and chemical effects that eventually induce cell death. C and D. Illustration of the interaction between focused ultrasound waves and gas bubbles at the different wave phases. The bubbles contract during the compression phase and expand during the rarefaction phase. In the case of stable cavitation (C), they repeat these cycles indefinitely and fluctuate around a relatively constant size. However, with inertial cavitation (D), the bubbles expand further during each rarefaction phase, exceeding their previous maximum size, continually expanding until reaching a critical size limit, at which pressure from the surrounding fluid surpasses the strength of the fluid-gas barrier causes bubble implosion.

Both stable and inertial cavitation elicit downstream impact within biological systems. Stable cavitation induces the oscillation of the surrounding fluid, resulting in "microstreaming," the transmission of mechanical stress onto nearby structures. This can disrupt phospholipid membranes and cell-to-cell junctions. These findings were initially observed in studies aiming to decipher the potential role of ultrasound as a disinfecting technology, and observations were made in bacterial and fungal pathogens [35,36]. In cancer cells, the incorporation of a sonosensitizer into the cellular membrane may cause destabilization that, in turn, sensitizes the cells to stable cavitation and microstreaming. This theory is described by Hiraoka *et al.*, who demonstrated the cytotoxic effects of ultrasound in hematoporphyrin-sensitized cells at intensities well below those that induce inertial cavitation [37]. Additionally, they observed that decreased hydrophobicity of the sonosensitizer was associated with enhanced sonomechanical effects. This suggests that the sonosensitizer's ability to integrate into the phospholipid bilayer is central to the efficacy of low-intensity ultrasound in this context [37]. Furthermore, Stepniewski *et al.* previously described and modeled this physical interaction between hematoporphyrin and lipid membranes [38].

Nevertheless, the more extensively studied cavitation-dependent mechanisms in SDT are thought to be related to the effects of inertial cavitation (Fig. 2). The implosion of gas bubbles as a result of inertial cavitation releases substantial energy in the form of heat, light, and shock waves for an ultrashort period (around 350 ps). Several studies have attempted to quantify the thermal energy released during bubble implosion, with varying reports estimating the temperatures reached to be 4300 K [39], 7000 K [40], or even as high as 20,000 K [41]. The extreme thermal energy released during bubble collapse results in a host of downstream effects. Furthermore, inertial cavitation-related forces are a transient phenomenon and cannot be sustained continuously, as microbubbles burst in the sonicated area and are pushed away by surrounding radiation forces, leading to temporary microbubble depletion in the target area. Replenishment of the depleted microbubbles depends on perfusion to the area and vessel type [42].

Additionally, sonoluminescence, the emission of light as a result of bubble collapse after inertial cavitation, has attracted wide interest as a potential therapeutic mechanism if better understood and directed. Early experiments describing the sonoluminescence phenomenon date as early as 1934, when Frenzel *et al.* documented the generation of light upon the exposure of distilled water, but not degassed water, to ultrasound [43]. A series of studies continued to solidify evidence on the involvement of sonoluminescence in the downstream effects of SDT. The evidence stemmed from experiments that detected sonoluminescence in a different setting that ranged from a sonicated agar gel phantom [44] to the *in vivo* setting with the aid of fluoresceinyl *Cypridina* luminescent analogue (FCLA) and a highly sensitive imaging system [44–46]. More recently, Datta *et al.* generated and validated a theoretical model of the hydrodynamics of bubble cavitation, bubble collapse, thermal energy release, and resulting sonoluminescence emission while using clinically relevant ultrasound parameters (frequencies between 200kHz and 1MHz, and pressures <10 MPa) for transcranial low-intensity FUS [47]. They also showed that stronger sonoluminescence can be generated from the use of lipid-coated oxygen microbubbles as compared to the FDA-approved Lumason™ microbubbles [47], suggesting the potential for the lipid-coated oxygen microbubbles to augment the efficacy of SDT. Ultimately, the generated sonoluminescence can be harnessed to generate oxidative stress via the excitation of sonosensitizers through a photochemical reaction [48,49]. Briefly, sonoluminescence can excite the sonosensitizer to a temporary singlet state followed by a triplet state. In its triplet state, the excited sonosensitizer can interact directly with oxygen to generate a singlet oxygen ($^1O_2$) or with biological molecules to indirectly generate ROS. This photo-dependent mechanism encouraged the repurposing of photosensitizers, such as porphyrin-derived agents, to be used in SDT. Herein, Umemura et al. investigated the use of hematoporphyrin as a sonosensitizer, and they supported the theory behind sonoluminescence-based photochemical effect through the detection of generated light. Additionally, they demonstrated the overlap between the sonoluminescence peak wavelength (400–150 nm) and the excitation peak for hematoporphyrin (411 nm) [50]. However, some studies debunk the theory that the production of ROS is reliant on sonoluminescence and photochemical reactions by showing that photoinactive agents, such as DPCH-PNa [51], could still induce ROS accumulation and cancer cell death when exposed to ultrasound. Hence, more robust mechanistic in vivo studies with real-time detection of sonoluminescence intensity and wavelength are necessary. Additionally, a deeper understanding of sonosensitizer excitation and continuous monitoring of ROS accumulation are needed. Such studies will help clarify the exact sequence of events and allow for more sophisticated closed-loop feedback systems for SDT. Finally, the third form of energy released from inertial cavitation is sonomechanical energy in the form of shock waves that occur after microjetting and bubble implosion [52]. The sonomechanical energy produced can contribute to cell membrane disruption with the assistance of sonosensitizer-mediated destabilization [53].

2.2. Intracellular molecular mechanisms

The generation of intracellular ROS and subsequent cell death is among the prevailing theoretical mechanisms regarding the antitumoral effects of SDT (Table 1) [17,54–58]. This mechanism posits that the ultrasound-mediated activation of a sonosensitive agent generates free radical species that enact macromolecular damage and eventually causes downstream cytotoxicity through cellular damage, mitochondrial dysfunction, and the activation of intrinsic apoptotic pathways [59–63].

Several reports have demonstrated the cytotoxic effects of ROS

Table 1
In vitro studies assessing SDT in glioma models. PW = pulsed wave, NR = not reported, DVDMS = sinoporphyrin sodium, 5-ALA = 5-aminolevulinic acid, HMME = hematoporphyrin monomethyl ether, LP = lipid nanoparticle, TMZ = temozolomide, BNPD-Ce6@Plt = boron nitride nanoparticles carrying chlorin e6 conjugated to platelets, ROS = reactive oxygen species.

Study	Cell line	Frequency	Power	Duty cycle	Time	Agent	Dose	Mechanisms investigated
An [92]	U118MG, U87MG	1.0 MHz	0.5 W/cm^2	100 %	1 min	DVDMS	0 – 8.0 µM	ROS mediated intrinsic apoptosis, anti-angiogenesis
Bilmin [59]	RG2	1.0 MHz	0.5–1.0 W/cm^2	40 %	3 min	5-ALA	10 µg/mL	Apoptosis
Chen [66]	U251	1.0 MHz	0.5 W/cm^2	NR	2 min	ZnPcS2P2	0.625–20 µg/mL	ROS mediated intrinsic apoptosis, cell membrane destruction
Chen [65]	C6	1.0 MHz	1 W/cm^2	NR	NR	HMME, TMZ	1.0 µg/mL each	ROS mediated intrinsic apoptosis
Dai [17]	C6	1.0 MHz	1 W/cm^2	PW	1 min	HMME	10 µg/mL	ROS mediated intrinsic apoptosis
Dai [87]	C6	0.5 MHz	1 W/cm^2	PW	1 min	HMME	10 µg/mL	ROS and Ca^{2+} mediated intrinsic apoptosis
Endo [85]	C6, U87MG	1.0 MHz	0.16 W/cm^2	50 %	1 min	5-ALA	200 µg/mL	Apoptosis
Hao [64]	C6	0.5 MHz	1.0 W/cm^2	PW	1 min	HMME	10 µg/mL	ROS and Ca^{2+} mediated intrinsic apoptosis
Hayashi [72]	U521MG, U105MG	0.95 MHz	0.3 W/cm^2	NR	5–30 sec	Porfimer sodium	75 µg/mL	NR
Ju [83]	SNB19, U251, U87MG	1.0 MHz	1 W/cm^2	20 %	2 min	5-ALA	0–1.0 mM	ROS mediated intrinsic apoptosis, hyperthermia
Li [60]	C6	1.0 MHz	0.5 W/cm^2	NR	2 min	HMME	10 µg/mL	ROS mediated intrinsic apoptosis
Li [61]	C6	1.0 MHz	0.5 W/cm^2	NR	2 min	HMME	10 µg/mL	ROS and Ca^{2+} mediated intrinsic apoptosis
Li [62]	C6	1.0 MHz	0.5 W/cm^2	NR	2 min	HMME	10 µg/mL	ROS mediated intrinsic apoptosis
Liu [177]	U87MG	0.5 MHz	0.5 W/cm^2	50 %	5 min	DVDMS-Mn-LPs	1 µg/mL DVDMS	ROS mediated intrinsic apoptosis
Pellegatta [122]	GL261	0.983 MHz	NR	10 %	20 min	Fluorescein	0.5 mg/mL	ROS mediated intrinsic apoptosis, immunomodulation
Pi [56]	U87MG	0.996 MHz	1.7 W/cm^2	30 %	1 min	DVDMS	10 µg/mL	ROS mediated intrinsic apoptosis
Shan [120]	GL261, CT2A	1.0 MHz	1 W/cm^2	100 %	2 min	Biomimetic nanoparticle (AMNP@J+C)	1 µg/mL JQ1 and 5 µg/mL Ce6	ROS mediated intrinsic apoptosis, immunomodulation
Sheehan [58]	C6, U87MG	1.1 MHz	10 W/cm^2	10 %	3 min	5-ALA	1 mM	ROS mediated intrinsic apoptosis
Shen [20]	U87MG	0.970 MHz	0.32 W	1–30 %	3 min	DVDMS	10 µg/mL	ROS mediated intrinsic apoptosis
Suehiro [74]	U251, U87MG	3.0 MHz	2 W/cm^2	20 %	3 min	5-ALA	1 mM	ROS mediated intrinsic apoptosis
Sun [55]	U373	1.0 MHz	0.45 W/cm^2	NR	1 min	DVDMS	0.5 µM - 10 µM	Cell membrane damage, ROS mediated intrinsic apoptosis, DNA damage
Sun [63]	C6	NR	NR	NR	NR	Liposomal DVDMS	1 µM	ROS mediated intrinsic apoptosis
Tserkovsky [15]	C6	0.88 MHz	0.2–0.7 W/cm^2	NR	NR	Photolon	1 µg/mL	NR
Wang [67]	GL261	1 MHz	1 W/cm^2	NR	30 sec	BNPD-Ce6@Plt	5:1 Plt:Gl261	ROS mediated intrinsic apoptosis, DNA damage, immunomodulation
Xu [70]	U251	1.0 MHz	0.5 W/cm^2	NR	1 min	Porfimer sodium	20 µg/mL	ROS mediated intrinsic apoptosis
Xu [71]	U251	1.0 MHz	0.5 W/cm^2	NR	2 min	Porfimer sodium	20 µg/mL	ROS mediated intrinsic apoptosis
Yoshida [89]	F98	0.22 MHz	20 W	100 %	4 min	5-ALA	200 µg/mL	ROS mediated intrinsic apoptosis, hyperthermia, anti-angiogenesis
Zhou [109]	LN229, GL261	1.0 MHz	1.0–2.0 W	20 %	5 min	TMZ, Liposomal TMZ	200–500 µM	ROS mediated intrinsic apoptosis, DNA damage, immunomodulation
Zhu [121]	C6	NR	1.0 W/cm^2	NR	2 min	NLG919 prodrug	1.8 mM	ROS mediated intrinsic apoptosis, immunomodulation

generated by SDT *in vitro* as well as *in vivo* [17,64–66]. Hao *et al*. studied SDT *in vitro* using the BBB permeable hematoporphyrin monomethyl ether (HMME) in rodent-origin glioma cells and observed intracellular calcium overload, decreased mitochondrial membrane potential, and cytochrome-c release. These findings suggest that HMME-SDT enhances ROS production by increasing intracellular calcium ion flux and activates apoptotic pathways [64]. Further supporting the influence on HMME-SDT on mitochondrial signaling, Dai *et al.* observed loss of mitochondrial membrane potentials coupled with increased levels of Bax, caspase-9, caspase-3, and decreased levels of BCL-2 and Fas-L after HMME-SDT treatment in the same rodent glioma *in vitro* model [17]. Chen *et al.* similarly observed cristae loss, vacuole formation, mitochondrial swelling, and apoptotic body formation with transmission electron microscopy in response to SDT combined with temozolomide administration in an *in vitro* C6 glioma model [65]. These studies supporting the generation of ROS in response to SDT are further corroborated by investigations demonstrating the loss of SDT efficacy with the administration of antioxidant agents. For example, pretreatment with N-acetylcysteine (NAC), a potent free radical scavenger, has been shown to increase cell viability after SDT significantly [55,67].

While ROS generation sites are mostly cytoplasmic, many SDT agents also exhibit subcellular localization, which may play a critical role in their cytotoxic activity. FUS, at parameters typically used to achieve SDT, can permeabilize the nuclear membrane. ROS generation can result in the expression of oxidative-stress-related genes, including Nrf2, NF-κB, and MAP kinases that propagate DNA damage [55,68]. Following SDT with sinoporphyrin sodium (DVDMS), Sun *et al.* observed ROS-mediated DNA fragmentation through double-stranded breaks and increased levels of cleaved caspase-3 in human U373 glioma cells [55]. Interestingly, Shen *et al*. similarly studied DVDMS in human glioma cells (U87MG) and found that the activation of DVDMS with FUS resulted in significant ROS generation as well as a loss of mitochondrial membrane potential. They observed that DVDMS tends to localize in the mitochondria subcellularly, the main target of the oxidative stress imparted by ROS-generation, which resulted in subsequent lipid peroxidation and mitochondrial membrane depolarization, activating the caspase apoptosis pathway [18,20,69].

Specifically, GBM is a notoriously heterogenous tumor, and, predictably, studies of different populations of glioma cells have shown variable effects of SDT. In a study in an immortalized GBM cell line (U251), Xu *et al.* observed that SDT with porfimer sodium was less effective in different cell populations, evidenced by decreased porfimer sodium uptake and, consequently, decreased ROS production [70,71]. Hayashi *et al*. similarly investigated porfimer sodium-SDT in both U251 and U105 malignant glioma cells, observing that the cell surface transporter LRP/α2MR is necessary for porfimer-sodium efficacy as its blockade, or absence as observed in U105 cells, mitigates the effects of the SDT. Notably, this study also observed the formation of several pores and dimple-like craters on cell surface membranes using scanning electron microscopy after porfimer-SDT, also suggesting a mechanism of action involving the disruption of cellular membranes as detailed earlier [72]. In an *in vitro* study of 5-aminolevulinic acid (5-ALA)-SDT in glioma cells, Shono *et al*. demonstrated that the efficacy of SDT was potentiated by administering celecoxib, which downregulates the multidrug-resistant protein MDR1, a protein implicated in GBM chemoresistance through the Akt/NF- κB pathway. Notably, they observed increased intracellular protoporphyrin IX (PpIX) accumulation as well as increased levels of cleaved caspase-9, caspase-3, and PARP, supporting the activation of an intracellular apoptotic pathway in response to SDT [73]. Importantly, these studies highlight the dependency of SDT efficacy on intracellular SDT agent accumulation to facilitate ROS-generation and caspase-dependent apoptotic pathway activation. Given the potential variability in the accumulation of these agents in different cell types and different brain regions, single-agent SDT with heme-pathway derivative agents may not sufficiently achieve therapeutic efficacy, and strategies to increase intracellular PpIX may be necessary [20,74]. Furthermore, given the relative hypoxia of GBM and its tumoral microenvironment, the efficiency of ROS generation as an antitumoral effect remains controversial and poorly understood [75]. Therefore, an important step in clinical translation is a better understanding of the pharmacokinetics of various SDT agents, including intratumoral sonosensitizer accumulation and BBB penetrance, and subsequent ROS-generation capacity.

2.3. *Pyrolysis and hyperthermy*

Pyrolytic reactions involve molecular decomposition in response to heat in the absence of oxygen. Pyrolysis, in the context of SDT, involves the thermal degradation of biological tissues as the application of FUS, with or without a sonosensitizer agent, can lead to localized heating within tissues [76,77]. Pyrolysis can contribute to the generation of ROS, through the generation of free radical from water molecules that are present in the vicinity of the cavitating bubble implosion, which can release high amounts of heat [78]. Similarly, pyrolysis can involve the sonosensitizer by exciting it into an intermediate state with subsequent free radical generation [78]. Peak focal intensity and the duration of sonication can affect the degree of hyperthermia achieved. As highlighted by Ohmura *et al.*, at a peak focal intensity of 25 W/cm [2] and 1.04 MHz frequency for 2 min, no significant temperature change was observed in a C6 rat glioma model [19]. However, with exposure to FUS at 110 W/cm^2 and 1.04 MHz frequency for 2 min, significant temperature changes up to 50°C were observed [19]. Also in a rat glioma model, Wu *et al.* observed only modest temperature increases in response to 20 min of FUS at 5.5 W/cm^2 and a 1.06 MHz frequency, from 32.3°C to 33.2°C and 37.2°C to 38.4°C, in groups with respective starting body temperatures of 32°C and 37°C [79]. Notably, both the 32°C and 37°C core body temperature groups demonstrated similarly improved survival in response to 5-ALA mediated SDT, relative to the 5-ALA and FUS alone. controls, suggesting temperature may not play a critical role in the efficacy of 5-ALA SDT [79].

While temperature increases alone can be inherently toxic to glioma cells, this thermal effect can also augment the cytotoxicity of sonosensitizers [80,81]. Kinoshita & Hynynen observed, in a HeLa cell model, that intracellular PpIX acted synergistically with hyperthermia to induce cellular killing [82]. These findings have been corroborated by investigation in SNB19 and U87MG glioma cells, showing that hyperthermotherapy results in increased ROS production, mitochondrial membrane potential loss, and apoptosis in response to 5-ALA SDT *in vitro.* [83] Additionally, localized elevations in temperature from inertial cavitation and bubble collapse can generate free radical products from a sonosensitizing molecule that enact indiscriminate macromolecular and cellular damage [84]. Temperature variations during sonication significantly influence the SDT response, suggesting that careful control of thermal parameters can help optimize and ensure consistent treatment outcomes [54].

Many *in vitro* investigations of SDT attempt to control for FUS's thermal effects by implementing measures to ensure temperature control of *in vitro* conditions (Table 2) [65,85–87]. For example, Chen *et al.* utilized a wet sponge applied to the bottom of culture plates to ensure temperature control in an assessment of the SDT capabilities of temozolomide, which showed efficacy despite the temperature-controlled experimental conditions [65]. Similarly, Endo *et al.* utilized a 37°C water bath to ensure temperature regulation in an *in vitro* assessment of the effects of various porphyrin-derivative SDT agents, demonstrating induction of apoptosis in the absence of meaningful temperature changes [85]. Given the evidence supporting SDT's efficacy *in vitro* and *in vivo* irrespective of temperature conditions and the typically non-ablative parameters utilized in low-intensity FUS as implemented in SDT, it is unlikely that significant hyperthermia is involved in the mechanism of SDT. However, temperature monitoring, as will be discussed further later, remains a critical aspect of ensuring consistent *in situ* FUS parameters.

Table 2

In vivo studies assessing SDT in glioma and selected other solid tumor models. ROA = route of administration, S = subcutaneous, IC = intracranial, NR = not reported, DVDMS = sinoporphyrin sodium, 5-ALA = 5-aminolevulinic acid, HMME = hematoporphyrin monomethyl ether, MB = microbubble, LP = lipid nanoparticle, TMZ = temozolomide, Ce6 = chlorin e6, BNPD = boron nitride nanoparticle, IV = intravenous, IP = intraperitoneal, PO = per os (orally), ROS = reactive oxygen species.

Study	Animal	Tumor location	Cell line	Frequency	Power	Duty cycle	Time	Agent	Dose	ROA	Mechanisms investigated
Glioma models											
An [92]	BALB/c nude mice	S	U118MG, U87MG	1.0 MHz	500 mW/cm^2	100 %	3 min	DVDMS	1.0–2.0 mg/kg	Injection	ROS mediated intrinsic apoptosis, anti-angiogenesis
Jeong [13]	Sprague-Dawley rats	IC	C6	1. MHz	2.65 W/cm^2	NR	20 min	5-ALA	60 mg/kg	IV	N/A
Ju [83]	BALB/c nude mice	S	SNB19, U251, U87MG	1.0 MHz	2 W/cm^2	NR	10 min	5-ALA	250 mg/kg	IP	ROS mediated intrinsic apoptosis, hyperthermia
Liu [177]	BALB/c nude mice	S, IC	U87MG	0.5 MHz	1.5 W/cm^2	NR	10 min	DVDMS-Mn-LPs	10 mg/kg DVDMS	IV	ROS mediated intrinsic apoptosis
Nonaka [81]	Wistar rats	IC	C6	1.04 MHz	25–250 W/cm^2	NR	10 sec - 5 min	Rose Bengal	50 mg/kg	IV	Hyperthermia
Ohmura [19]	Wistar rats	IC	C6	1.04 MHz	10–25 W/cm^2	NR	5 min	5-ALA	100 mg/kg	PO	N/A
Pellegatta [122]	C57Bl/6 mice	IC	GL261	0.485 MHz	NR	10 %	20 min	Fluorescein	10 mg/kg	IP	ROS mediated intrinsic apoptosis, immunomodulation
Pi [56]	BALB/c nude mice	IC	U87MG	0.996 MHz	1.7 W/cm^2	1 %	1 min	DVDMS	2 mg/kg	IV	ROS mediated intrinsic apoptosis
Prada [57]	Sprague–Dawley rats	S	C6	0.35 MHz	2–6 W/cm^2	10 %	NR	Fluorescein	10 mg/kg	IP	ROS mediated intrinsic apoptosis, DNA damage
Shan [120]	C57Bl/6 mice	IC	GL261	3.0 MHz	1 W/cm^2	100 %	6 min	Biomimetic nanoparticle (AMNP@J+C)	2 mg JQ1 equiv./kg, 10 mg Ce6 equiv./kg	IV	ROS mediated intrinsic apoptosis, immunomodulation
Shono [73]	C57Bl/6 mice	IC	GSCs	1.0 MHz	2 W/cm^2	NR	2 min	5-ALA	200 mg/mL	IP	ROS mediated intrinsic apoptosis
Song [93]	Wistar rats	IC	C6	1.0 MHz	0.5 W/cm^2	NR	2 min	HMME	10 ng/kg	IV	Apoptosis, anti-angiogenesis
Suehiro [74]	BALB/c nude mice	IC	U251, U87MG	2.2 MHz	0.5 kW/cm^2	100 %	5 min	5-ALA	100 mg/kg	PO	ROS mediated intrinsic apoptosis
Sun [93]	BALB/c nude mice	S, IC	C6	1.0 MHz	1 W	100 %	1 min	Liposomal DVDMS	1 mg/kg	IV	ROS mediated intrinsic apoptosis
Tserkovsky [16]	Random-bred rats	S	C6	1 MHz	0.4–1.0 W/cm^2	NR	10 min	Photolon	2.5 mg/kg	IV	N/A
Wang [67]	BALB/c nude and C57Bl/6 mice	IC	GL261	1 MHz	1 W/cm^2	NR	4 min	Ce6, BNPD-Ce6	0.0175 mg/kg	IV	ROS mediated intrinsic apoptosis, DNA damage, Immunomodulation
Wu [79]	Sprague-Dawley rats	IC	C6	1.06 MHz	5.5 W/cm^2	NR	20 min	5-ALA	60 mg/kg	IV	Apoptosis, hyperthermia
Yoshida [86]	Fisher rats	IC	F98	0.22 MHz	18 W	100 %	30 s	5-ALA	100 mg/kg	IP	ROS mediated intrinsic apoptosis, hyperthermia, anti-angiogenesis
Zhou [109]	Mice (unspecified)	IC	LN229, GL261	NR	NR	NR	NR	TMZ, Liposomal TMZ	NR	NR	ROS mediated intrinsic apoptosis, DNA damage, immunomodulation
Zhu [121]	BALB/c nude mice	IC	C6	NR	1.0 W/cm^2	NR	10 min	NLG919 prodrug	5 mg/mL	IC hydrogel	ROS mediated intrinsic apoptosis, immunomodulation
Other solid tumor models											
Chen [117]	BALB/c nude mice	S	4T1 (breast)	1 MHz	0.5 W/cm^2	50 %	10 min	LY364947	0.43 mg/kg	IV	ROS mediated intrinsic apoptosis, Immunomodulation
Gao [88]	BALB/c nude mice	S	HUVEC, SAS (tongue)	1.1 MHz	2 W/cm^2	50 %	5 min	5-ALA	250 mg/kg	IP	Anti-angiogenesis
Li [18]	BALB/c nude mice	S	Hep-G2 (liver)	1.1 MHz	1 W/cm^2	10 %	10 min	DVDMS	2 mg/kg	IV	ROS mediated intrinsic apoptosis
Nesbitt [90]	SCID mice	S	MIA PaCa−2 (pancreas)	1 MHz	3.5 W/cm^2	30 %	7 min	O_2-MB-Gemcitabine	0.464 mg/kg Gemcitabine	IV	ROS mediated intrinsic apoptosis
Wang [112]	BALB/c nude and BALB/c mice	S	B16F10 (melanoma)	1.1 MHz	2 W/cm^2	10 %	5 min	5-ALA	250 mg/kg	IP	Immunomodulation

2.4. Anti-angiogenesis and hypoxia

In addition to the ability to induce cancer cell death via the mechanisms discussed above, SDT also possesses the capacity to elicit anticancer effects through the modulation of the tumor microenvironment and non-cancer cell populations. Although investigations of this phenomenon in GBM models are scarce, analogous evidence from other solid tumor models provides support for the role of antiangiogenic processes in the mechanistic activity of SDT.

Herein, Gao *et al.* first described the antiangiogenic properties of SDT through their *in vitro* and *in vivo* studies on primary human umbilical vein endothelial cells (HUVECs) [88]. This group showed that 5-ALA--based SDT significantly attenuated the proliferation, migration, invasion, and tube formation capacity of HUVECs *in vitro* [88], phenomena that are central to the process of angiogenesis and vascularization of new regions. They have further validated these observations *in vivo* by showing that 5-ALA-based SDT substantially decreased microvessel density, attenuated vascular endothelial growth factor (VEGF) secretion, and altered vascular ultrastructure within human tongue cancer (SAS) tumors xenografted in mice [88]. Furthermore, several studies using different sonosensitizers have also confirmed the impact of SDT on angiogenesis by showing a significant reduction in VEGF expression after treatment in breast cancer [89], pancreatic cancer [90], and small cell lung cancer *in vivo* and *in vitro* models [91]. In studies within GBM models, An *et al.* observed significant vascular obstruction in tumors in response to SDT with DVDMS [92]. Song *et al.* additionally studied the effects of HMME SDT on angiogenesis in a C6 glioma model. Their results demonstrated inhibited glioma angiogenesis, as evidenced by decreased microvessel density and VEGF levels, in response to HMME + SDT, relative to either intervention alone, at 24 h and 3 days, with normalization to control levels by 7 days post-treatment [93]. Similarly, using 5-ALA/protoporhyrin IX (PpIX) in an F98 glioma model, Yoshida *et al.* observed decreased angiogenesis at the tumor margin in response to 5-ALA-based SDT [86].

On the other hand, the hypoxic microenvironment of some tumors has been shown to impact the efficacy of SDT. In specific, GBM is known to have a distinctively hypoxic environment with overexpression of hypoxia inducible factors that further promote malignant behavior [94, 95]. It is well established that the presence of oxygen helps mediate the ROS-related effects of SDT to promote cell death [96]. The hypoxic environment present within many solid tumors may hinder these mechanisms and reduce SDT efficacy in hypoxic regions. As such, strategies to mitigate the hypoxia inherent to many tumor microenvironments have gained attention to augment the efficacy of SDT. Such strategies include increasing oxygen abundance by delivering oxygen using nanocarriers [97], inhibiting oxygen utilization for cellular respiration [98], catalyzing the endogenous production of oxygen within the cells [99–101], or a combination of these processes [96]. For example, Zhang *et al* [96]. utilized manganese dioxide (MnO_2)-based nanoparticles to carry both a sonosensitizer (indocyanine green) and metformin, an oxidative phosphorylation inhibitor. In this paradigm, metformin acts to attenuate the consumption of oxygen by the cancer cells, and MnO_2 catalyzes the breakdown of H_2O_2 into oxygen. The application of this treatment strategy improved tumor oxygenation *in vivo* and augmented the efficacy of SDT in a breast cancer model [96]. A similar paradigm was followed to engineer a sonosensitizer composed of nanozyme PLGA particles linked to angiopep-2, to enhance BBB permeability and improve glioma targeting, and loaded with MnO_2 to deplete H_2O_2 and exhaust glutathione and IR780. The results showed that the tested particles augmented the efficacy of SDT in a GBM model by targeting distribution, attenuating hypoxia, and subsequently inducing increased ROS accumulation [102].

2.5. Immunomodulation

SDT has been shown to potently activate antitumoral immunity by inducing immunogenic cell death (ICD) and transforming immunologically "cold" tumors to immunologically activated, 'hot' tumors – a particularly advantageous property for tumors such as GBM that exhibit diverse and potent immunosuppressive characteristics [103–105]. The immune activation induced by SDT may play a critical role in its therapeutic efficacy, primarily through mechanisms involving the release of damage-associated molecular patterns (DAMPs) and subsequent lymphocyte activation as a result of ultrasonic cavitation (Fig. 3) [106, 107]. Both foundational and clinical studies have provided robust evidence that SDT can induce ICD, characterized by the release of key DAMPs such as calreticulin (CRT), extracellular adenosine triphosphate (ATP), and high-mobility group box 1 (HMGB1) as well as surface localization of heat shock proteins (HSPs) [108–110]. Intracerebrally, these molecules can activate microglial and dendritic cells and initiate a sequence of immune responses that target tumor cells and amplify DAMP release, creating a synergistic anti-tumor cascade with SDT. Additionally, evidence in other solid tumors supports the ability of SDT to induce an abscopal effect, whereby clonal expansion of activated T-cells results in regression of remote metastatic disease [111]. While this effect is less likely to be observed in GBM clinically, given its lack of propensity for peripheral metastasis, these studies nonetheless support the immune-priming effects of ICD secondary to SDT.

The immunogenic potential of SDT has been studied across various cancers and agents, with nanomaterial-based SDT agents demonstrating particular utility in their immune-modulating capacity when applied in SDT. In a melanoma xenograft model, Wang *et al.* demonstrated that SDT with 5-ALA activates macrophages and dendritic cells, evidenced by increased levels of dendritic cell markers and resulted in M2-like (immunosuppressive) to M1-like (pro-inflammatory) macrophage polarization shifts [112]. Song *et al.* investigated a dual-target approach combining SDT and immunotherapy demonstrated a synergistic effect in a lung metastasis and recurrence model [113]. The combination treatment resulted in increased infiltration of CD8 + T cells, leveraging a chimeric protein for SDT alongside a checkpoint inhibitor for immunotherapy. Wang *et al.* studied a formulation using chlorin e6 and iron oxide nanoparticles (Ce6/Fe3O4) in SDT, and found this technique promoted the uncloaking of endogenous tumor neoantigens in a murine model of melanoma [114]. When combined with an immune checkpoint inhibitor, this approach elicited a robust CD8 + cytotoxic T cell response, ultimately resulting in the complete regression of the primary tumor. Similarly, Um *et al.* developed a nanomedicine-based SDT approach in a colon cancer model that facilitated the release of DAMPs through cell membrane disintegration, leading to significantly enhanced antitumoral immunity *in vitro* and *in vivo* [115]. Si *et al.* demonstrated that treatment with liposome-encapsulated perfluorohexane (LIP-PFH) nanoparticles, activated using low-intensity focused ultrasound (LIFU), successfully suppressed the growth and triggered apoptosis of 4T1 breast cancer cells by producing ROS in both *in vitro* and *in vivo* conditions. Furthermore, nanoparticle-based SDT was found to induce ICD in breast cancer cells, promoting dendritic cell maturation and enhancing the population of cytotoxic T cells [116]. Chen *et al.* utilized a combination of decitabine, and LY364947 encapsulated in polymeric micelles and activated by FUS in a murine model of triple-negative breast cancer [117]. The treatment elicited a strong proinflammatory macrophage response and significantly increased $CD3^+$ $CD8^+$ T cell infiltration. Moreover, the combination therapy effectively prevented long-term recurrence, with evidence for durable immunity in the spleen displaying diverse memory T cell populations. In addition to studies demonstrating the potential for ICD induction, several reports have explored modulation of the tumor microenvironment (TME) through nanomedicine-based SDT agents, including facilitating the release of tumor-associated antigens, and modulating oxygenation of the TME [116,118,119].

Unfortunately, immune responses in GBM models with SDT therapy remain relatively underexplored. One study examined the sonotoxic effects of temozolomide in two immortalized GBM cell lines *in vivo*

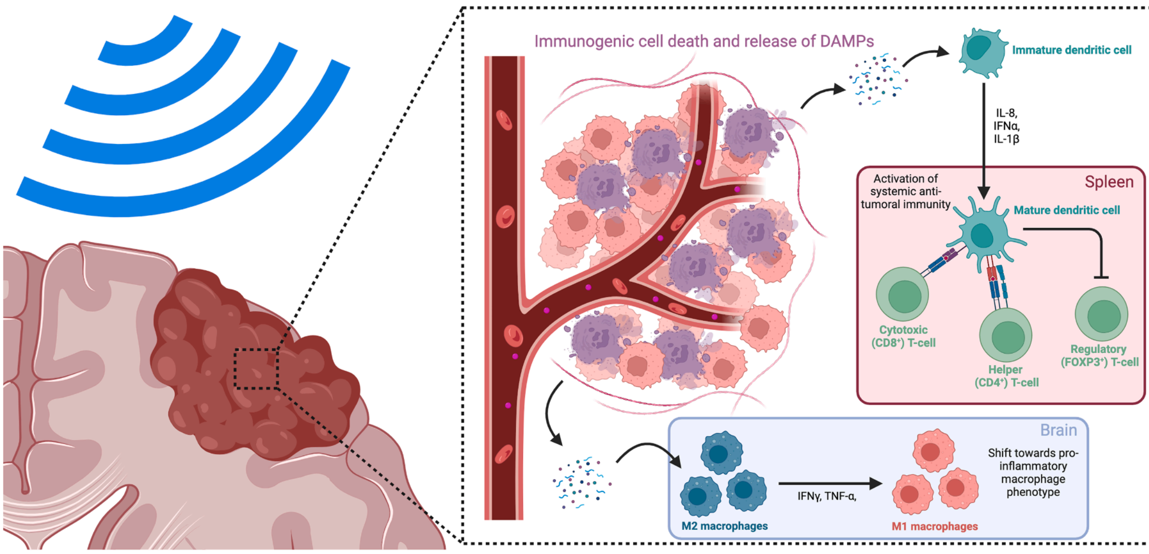

Fig. 3. SDT modulates the tumor immune response. SDT can result in immunogenic cell death, causing the release of damage-associated molecular patterns and pro-inflammatory cytokines, including IL-8, IFNα, and IL- 1β. These signals result in the activation of dendritic cells, which can activate systemic immune responses by increasing levels of cytotoxic and helper T-cells, as well as downregulating regulatory T-cells. Additionally, the presence of cytokines, including IFNγ and TNFα results in polarization of macrophages from the M2 (immunosuppressive) to the M1 (pro-inflammatory) phenotype.

[109]. They showed that combined temozolomide and FUS treatment enhanced the M1 macrophage response. Additionally, the local immune microenvironment showed significantly elevated levels of cytokines IL-6, IL-8, and IFN-γ. Another study by Shan *et al.* explored immune responses in an orthotopic GL261 mouse model treated with AMNP@J + C (a biomimetic nanoparticle containing a ROS responsive component co-encapsulated with the sonosensitizer Ce6 and bromodomain-containing protein 4 inhibitor JQ1) activated by FUS [120]. They found that this approach led to the highest levels of tumor necrosis factor alpha (TNF-α), interleukin 6 (IL-6), and interferon gamma (IFN-γ) in peripheral blood serum 72 h post-injection, compared to a PBS control group. Importantly, the AMNP@J + C treatment without ultrasound showed lower levels of these cytokines, highlighting the role of FUS stimulation in enhancing anti-tumor immune responses. Wang et al. utilized the GL261 cell line to investigate boron nitride nanoparticles loaded with Ce6 in combination with SDT [67]. The treatment not only induced cell death but also triggered a quasi-immune response characterized by an increase in CD80, CD86, and MHC-II expression on macrophages, promoting a shift toward the M1 macrophage phenotype. Zhu *et al.* developed an injectable hydrogel delivery system for GBM treatment that combined SDT and immunomodulation using hydrogel containing semiconducting polymer nanoparticles (SPNs) and the immune checkpoint inhibitor NLG919 prodrug linked by singlet oxygen ($^1O_2$)-cleavable linkers [121]. Upon sonication, SPNs produce $^1O_2$ for SDT, inducing ICD and activating the NLG919 prodrug, which, in turn, amplified antitumor immunity. While immunomodulatory effects of SDT are elicited in the tumor and its microenvironment they can also be detected peripherally. Pellegatta *et al.* studied an orthotopic GL261 mouse model treated with fluorescein mediated SDT, depletion of myeloid-derived suppressor cells (MDSCs) and concomitant robust infiltration of CD8 + T cells were observed locally in the SDT treated mice. They also observed a high circulating lymphocyte-to-monocyte ratio and a very low proportion of MDSCs peripherally, which they associated with better survival in SDT-treated mice [122].

Overall, these studies suggest that SDT can promote ICD and potentially work in combination with immune modulators to augment therapeutic response. These findings highlight the potential of SDT to act as a dual therapeutic strategy by directly affecting tumor cells and simultaneously modulating the immune system to fight cancer more effectively, a multimodal approach which may be particularly effective in GBM, a heterogenous tumor in a traditionally immune privileged site.

3. Real-time in-situ FUS monitoring

Real-time parameter monitoring is an essential aspect of clinical translation to account for inter-individual variability and precise dosing. This principle is implemented in existing neurologic interventions, such as individual cortical motor threshold determination in transcranial magnetic stimulation to ensure optimal stimulus intensity [123]. Similarly, monitoring *in situ* ultrasound effects and adjusting acoustic parameters in real-time is crucial to developing thermal and mechanical (e.g. microbubble-enhanced) ultrasound therapies [124–126]. Such real-time monitoring and adjustments improve precision, accuracy, safety, and enable dose-responses correlations of a given treatment. In the context of intracranial SDT, acoustic pressure or peak negative pressure represent critical parameters to monitor to ensure consistency of FUS intensity [127]. Furthermore, cavitation monitoring can detect stable versus inertial cavitation to ensure optimal cavitation levels to ensure desired effects [127]. These monitoring techniques are discussed in further detail below. To generate a closed-loop workflow, FUS parameters including the duty cycle, pulse repetition frequency, and sonication duration can be modulated based on real-time feedback to ensure safety as well as consistent inter-individual in-situ effects.

Furthermore, the ability to tune sonication based on knowledge of the actual mechanisms of action for SDT is predicated on the ability to monitor the sonication effects in physiologically relevant clinical conditions, and adjust ultrasound parameters to achieve the desired effects. Importantly, *in vitro* and *in vivo* experimental conditions may differ significantly given variations in skull and tissue characteristics and dimensions. Three techniques to monitor ultrasound effects *in vivo* are described: magnetic resonance (MR) thermometry (MRT), MR-acoustic radiation force imagine (MR-ARFI), and acoustic emissions monitoring (AEM) (Table 3).

3.1. MR thermometry

Although SDT does not traditionally elicit significant temperature increases, FUS can induce temperature changes within biological tissues, for which MRT is highly sensitive [128,129]. Typical applications

Table 3
Advantages and disadvantages of commonly implemented FUS monitoring techniques. MR = magnetic resonance, MR-ARFI = magnetic resonance acoustic radiation force imaging, FUS = focused ultrasound.

Technique	Advantages	Disadvantages
MR Thermometry	• Real-time temperature monitoring • High spatial resolution	• Limited sensitivity in low-temperature changes • Susceptible to motion artifacts and magnetic field inhomogeneities • Temporal resolution may not be sufficient for rapid FUS-induced changes.
MR-ARFI	• Direct measurement of FUS focal spot • Applicable to non-thermal FUS applications	• Limited to pretreatment verification, current applications limited in ability to continuously monitor during therapy. • Low sensitivity in soft tissue • Relatively slow acquisition time limits real-time feedback for dynamic processes such as cavitation, differential powering
Acoustic emissions monitoring	• Direct assessment of cavitation activity • Real-time monitoring allowing dynamic feedback on treatment progress and treatment optimization	• Limited spatial resolution • Susceptible to noise and artifacts from background signals originating from tissue and bubbles, which may interfere with imaging

of MRT in intracranial FUS rely on phase-shift in the proton resonance frequency (PRF); the relatively rapid acquisition times and high sensitivity for temperature changes in tissues make this technique particularly well-suited for *in situ* monitoring [130]. The ability to quantify temperature distributions and changes in real-time during intracranial FUS enhances treatment accuracy. MRT also facilitates dynamic adjustment of ultrasound parameters during treatment to promote desired therapeutic effects while minimizing off-target damage to surrounding tissues [131–134]. During high-intensity focused ultrasound (HIFU) MRT is a crucial aspect of monitoring thermal tissue damage to assess dosimetry in a thermometry-based feedback loop [130]. Recent advancements in MRT techniques have focused on improving spatial and temporal resolution, which are vital for effective treatment control. For instance, novel spiral-based volumetric acquisition methods can enhance the quality of temperature imaging, allowing for better visualization of the heated volumes during FUS procedures [132,135]. Additionally, motion-robust MRT techniques have been introduced to address the challenges posed by physiological movements, which can affect the accuracy of temperature measurements during treatment [135]. These innovations are particularly important in clinical applications, where patient movement can cause imaging artifacts, significantly impacting treatment efficacy.

As previously discussed, temperature changes induced by FUS parameters typically employed in SDT are relatively modest and of limited benefit [68–86]. PRF-based MRT demonstrates high spatial and temporal resolution and can detect temperature changes on the scale of 0.3°C for 1 cm^3 voxel sizes and generate temperature maps within seconds [128,136]. Clinically used imaging systems are precise to within 1° to 2°C and exhibit a temporal resolution of approximately 3.5 s [137]. In a rodent glioma model, Wu *et al.* applied MRT to monitor temperature increases in groups receiving FUS alone and 5-ALA SDT, observing temperature increase 0.9°C to 1.2°C [79]. Yoshida *et al.* also implemented MRT to ensure temperature control in SDT investigations *in vitro*. [86] However, the nature of thermal responses associated with SDT are often subtle and may not reach the levels typically monitored by MR thermography, which is more effective in scenarios involving significant thermal changes [128,131,137,138]. In SDT applications, where the goal is to induce localized cellular responses rather than large-scale thermal damage, the temperature changes may fall below the detection threshold of MRT, rendering MRT a less optimal choice for monitoring purposes. Although the development of methodologies to correct for magnetic field drift and other artifacts in MRT can enhance the reliability of MRT in clinical settings and make it a robust tool for monitoring ablative FUS dosimetry, given the lack of significant thermal effects induced by SDT, MRT may not represent an optimal monitoring modality in this application [139].

3.2. Acoustic emissions monitoring

AEM is a technique that detects the emitted frequency spectra generated in response to ultrasound waves generating stable and/or inertial cavitation, as previously discussed [140–142]. AEM was initially developed in the 1990s to perform contrast enhanced ultrasound (CEUS) imaging to enhance Doppler signal and assess tissue perfusion, relying on the emission of the harmonic signals generated by microbubbles cavitation. To this end, dedicated contrast specific algorithms were developed to visualize real-time circulating microbubbles separated from tissue signals [143]. Real-time microbubble linear oscillations in the brain have been detected with CEUS intraoperatively, facilitating the understanding of their distribution in different structures and lesions [144]. A form of transcranial AEM, that captures the spatial distribution and intensity of cavitation activity, passive cavitation imaging (PCI), provides information about the location and dynamics of the cavitation process [145–147]. Although relatively less studied in clinical contexts than MRT, this spatial information can be used to optimize treatment parameters and localize cavitation effects to targeted areas while avoiding unintended damage to adjacent tissues [148,149].

In their investigation of transcranial ablative FUS in four rhesus macaques, Arvanitis *et al.* utilized passive cavitation detection to monitor the safety and potential toxicity of FUS, concluding that the FUS exposure level should lie close to the internal cavitation threshold [150]. Other preclinical investigations have demonstrated high-spatial resolution techniques for cavitation monitoring. Notably, a 128-element array constructed by O'Reilly *et al.* was capable of achieving 1.25 mm lateral and 2 mm axial resolution in an *ex vivo* human skull model, a resolution comparable to clinically used MRT systems [151]. The integration of AEM cavitation imaging with FUS allows for real-time monitoring of cavitation dynamics, which can allow real-time ultrasound parameter monitoring. For instance, studies by Haworth *et al.* and Smith *et al.* have demonstrated that PCI can be used to classify the type and intensity of cavitation, correlating these parameters with therapeutic outcomes [147,149]. Additionally, *ex vivo* evidence in a hemolysis model using horse red blood cells suggests that cavitation energy is the most significant factor in determining the cellular safety of FUS [149].

By employing closed-loop feedback control systems that utilize AEM data, ultrasound parameters can be adjusted dynamically to maintain optimal cavitation conditions and ensure consistent ultrasound dosimetry *in situ.* [146,152] Furthermore, MR-integrated and compatible ultrasound systems have been developed to simultaneously monitor temperature and cavitation activity during FUS therapies, allowing for a more thorough assessment of treatment efficacy and safety [148,153]. This multimodal approach can help in validating models of cavitation-enhanced heating and in planning and guiding SDT procedures more effectively.

3.3. MR-ARFI

Acoustic radiation force (ARF) refers to the transfer of momentum that occurs whenever an acoustic wave is interrupted by a target, resulting in the absorption or reflection of energy [154]. These interactions result in transient and minimal displacement of tissue, directly related to the location, size, and shape of the focal spot of the ultrasound beam. The displacement is also dependent on the intensity of the ARF and the intrinsic elasticity of the tissue itself [154]. Hence, the ability to detect the ARF-mediated displacement in physiological conditions has garnered wide interest due to the ability to act as a feedback mechanism that provides information on the intensity of ultrasound waves at a specified location and to ensure precise targeting of areas of interest and ultimately safety and efficacy. Magnetic resonance imaging has the ability to apply pulse sequences that can detect the micro-metric changes in tissue position [155]. Accordingly, the use of MR-ARFI, which is the coupling of FUS pulses with MR motion-encoding gradients, can broaden the potential of both technologies and open new avenues for precise localization and non-invasive monitoring.

Indeed, several studies have investigated the ability of MR-ARFI to achieve precise focal spot localization *in vivo* in rodents [156–159] and larger animal models, including sheep [160,161] and nonhuman primates [161–167]. Additionally, *ex vivo* studies have also been carried out using human skulls to confirm the feasibility of using MR-ARFI technology and the potential accuracy in human contexts [138,168]. Recently, Mohammadjavadi *et al.* concluded a first-in-human trial investigating the feasibility and utility of MR-ARFI for transcranial FUS targeting (results are still in pre-print) [169]. Another application of MR-ARFI is in optimizing FUS delivery by correcting for aberrations that occur as ultrasound waves cross from one medium of propagation to another (i.e., crossing through the skull into much softer brain tissues) [170–172]. In this context, Hertzberg *et al.* used pigs transplanted with human skulls to demonstrate the ability of MR-ARFI to account for any displacement or changes caused by the transmission of FUS waves through the human skull [173]. Similarly, Marsac *et al.* demonstrated the benefit of using MR-ARFI for optimizing FUS delivery using human cadavers [174]. These studies have shown that MR-ARFI offers several advantages over computed tomography (CT)-based aberration correction, which is the currently used strategy during FUS procedures. These advantages include a greater margin for displacement correction and a near-optimal focus with MR-ARFI, possibly as the latter uses the same platforms as MR guided FUS (MRgFUS), which results in a lower risk for misalignment or misregistration [173].

Importantly, MR-ARFI offers the potential to fill the gap in our ability to monitor acoustic pressure and interrogate the efficacy of SDT or other low-intensity, non-thermal FUS applications in real-time. This is due to the direct correlation between the observed displacement of the target tissue and the ARF signal at that location, as well as the correlation between the ARF and the acoustic pressure [156]. Indeed, the displacement detected using MR-ARFI correlates with the degree of neurophysiological impact elicited by ultrasound stimulation [160]. Recently, Li *et al.* have shown that the accuracy of MR-ARFI in measuring acoustic intensity can be improved if the prediction model takes into consideration the viscoelasticity of the target tissue [175]. Therefore, integrating MR elastography into the setup could be considered to create individualized prediction models that accurately reflect acoustic pressure and, ultimately, the mechanical effects contributing to SDT.

The incorporation of MR-ARFI into clinical SDT workflows used for the transcranial delivery of low-intensity FUS offers several advantages, such as a non-invasive and reliable technique for localization, skull-induced aberration correction, and acoustic intensity monitoring. Of note, MR-ARFI functions independent of detectable increases in tissue temperature required for MRT, and also offers greater accuracy and flexibility in the case of integrated FUS-MRI platforms.

3.4. Closed-loop feedback for SDT in GBM

Despite various efforts to develop closed-loop feedback systems for FUS monitoring and parameter optimization, their integration into the SDT workflow remains immature and scarce in the literature. Few in vivo SDT studies have been conducted with ROS monitoring via chemiluminescence [176,177] or sonosensitizer accumulation monitoring via dynamic positron emission tomography [178]. Translation of these probes remains a major challenge, and the real-time monitoring of ROS generation in-situ is far from being a clinical reality. Similarly, despite the fact that MR-ARFI has been extensively studied as a technique to improve the localization and targeting accuracy of FUS, studies investigating its use in the context of SDT are lacking. Several unique challenges uniquely exist in the intracranial context, including varying tissue compositions, eloquent brain areas, and the skull, which acts as a physical barrier to intracranial propagation of acoustic energy [179].

Given the deficit in closed-loop feedback mechanisms in current SDT clinical trials for GBM (Summarized in Table 4), there is a critical need for studies involving the previously discussed monitoring approaches in the context of this disease. The subsequent integration of these closed-loop feedback systems into clinical devices and clinical trial plans should be a priority.

4. Conclusion

SDT is an emerging therapy for GBM and other solid tumors with significant clinical promise. Uncovering and detailing the mechanisms of its action require further investigation to be completely elucidated and linked to specific FUS parameters. Intracellular ROS generation coupled with other potentially high impact effects stemming from FUS and sonosensitizer activation have the potential to transform cancer therapy., Examples of mechanisms for accomplishing these effects include acoustic cavitation, immunogenicity, and antiangiogenic effects, among others. Real-time monitoring of thermal, mechanical, and cavitation-based effects with real-time adjustments of acoustic parameters to achieve the desired bio-effects, therapeutic responses, and repeatable, standardized treatments will accelerate clinically meaningful SDT applications.

Abbreviations

5-ALA: 5-aminolevulinic acid; AEM: acoustic emissions monitoring; ARF: acoustic radiation force; ATP: adenosine triphosphate; BBB: blood-brain barrier; BNPD: boron nitride nanoparticle; Ce6: chlorin E6; CEUS: contrast enhanved ultrasound; CNS: central nervous system; CRT: calreticulin; CT: computed tomography; DAMP: damage-associated molecular pattern; DVDMS: sinoporphyrin sodium; FCLA: fluoresceinyl Cypridina luminescent analogue; FUS: focused ultrasound; GBM: glioblastoma; HIFU: high-intensity focused ultrasound; HMME: hematoporphyrin monomethyl ether; HMGB1: high-mobility group box 1; HSP: heat shock protein; HUVEC: human umbilical vein endothelial cells; ICD: immunogenic cell death; IFN-γ: interferon gamma; IL-6: interleukin 6; LIFU: low-intensity focused ultrasound; LIP-PFH: liposome-encapsulated perfluorohexane; MR: magnetic resonance; MR-ARFI: magnetic resonance acoustic radiation force imaging; MRgFUS: magnetic resonance guided focused ultrasound; MRT: magnetic resonance thermometry; NAC: N-acetylcysteine; PCI: passive cavitation imaging; PDT: photodynamic therapy; PpIX: protoporphyrin IX; PRF: proton resonance frequency; ROS: reactive oxygen species; SDT: sonodynamic therapy; SPN: semiconducting polymer nanoparticle; TME: tumor microenvironment; TNF-α: tumor necrosis factor alpha; VEGF: vascular endothelial growth factor

CRediT authorship contribution statement

Harshal Shah: Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Hasan Slika:** Writing – original draft, Validation, Resources, Methodology, Investigation, Conceptualization. **Fnu Ruchika:** Writing – original draft, Resources, Methodology, Investigation, Conceptualization. **Danielle Golub:** Writing – original draft, Resources, Investigation, Data curation. **Michael Schulder:** Writing – review & editing, Supervision, Resources. **Henry Brem:** Writing – review & editing, Supervision, Resources. **Amir Manbachi:** Writing – review & editing, Supervision, Resources. **Jordina Rincon-Torroella:** Writing – review & editing, Validation, Supervision. **Chetan**

Table 4
Summary of concluded or ongoing clinical trials investigating SDT in GBM. 5-ALA= 5-aminolevulinic acid; GBM= Glioblastoma; HCl, hydrochloride; SDT, sonodynamic therapy.

Clinical Trial Identifier	Type	Population	Intervention	Findings	Reference
ChiCTR2200065992	Phase 1	Patients with recurrent glioblastoma	Intravenous systemic administration or per cutaneous arterial catheterization was used for local administration of hematoporphyrin with low-frequency focused ultrasound delivered via DK−102C instrument and combined with temozolomide	1/9 patients had stable disease. Median overall survival was 202.5 days. None of the nine patients demonstarted any adverse effects (neurological, hematological, or dermatological) related to SDT.	[180]
NCT04559685	Phase 0/1, open-label, first-in-human	Patients with recurrent high-grade glioma	Intravenous 5-aminolevulinic acid (SONALA−001) with MR-guided focused ultrasound (Exablate 4000 Type 2)	Recruiting, no published results	Clinicaltrials.gov
NCT04845919	Phase 2, non-randomized, single-arm	Patients with newly diagnosed cerebral glioblastoma	Oral 5-aminolevulinic acid (SONALA−001) with MR-guided focused ultrasound (Exablate 4000 Type 2)	Completed, no published results	Clinicaltrials.gov
NCT05362409	Phase 1, multicenter, open-label	Patients with recurrent high-grade glioma	Oral 5-ALA HCl with the CV01 ultrasound delivery device	Completed, no published results	Clinicaltrials.gov
NCT05370508	Phase 1/2 multicenter, open-label, dose escalation and expansion	Patients with progressive or recurrent GBM	Intravenous 5-aminolevulinic acid (SONALA−001) with MR-guided focused ultrasound (Exablate 4000 Type 2)	Terminated (due to funding issues and not due to safety concerns), no published results	Clinicaltrials.gov
NCT06039709	Phase 1	Patients with recurrent glioblastoma	Oral 5-ALA with neuronavigation-guided low-intensity focused ultrasound	Recruiting, no published results	Clinicaltrials.gov
NCT06665724	Phase 1, single center	Patients with newly diagnosed high-grade glioma	Oral 5-ALA HCl with the CV01 ultrasound delivery device	Recruiting, no published results	Clinicaltrials.gov

Bettegowda: Writing – review & editing, Validation, Supervision. **Pavlos Anastasiadis:** Writing – review & editing, Resources, Investigation. **Francesco Prada:** Writing – review & editing, Validation, Resources. **Graeme Woodworth:** Writing – review & editing, Validation, Supervision, Resources. **Betty Tyler:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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